

APLIKACJE WEBOWE

Reading the Docs, Writing the Errors

L1



by ***Mateusz Węglewski***



"Error 500. Something went wrong."

Co dokładnie poszło nie tak? Nie wiadomo.

Co ma teraz zrobić użytkownik? Nie wiadomo.

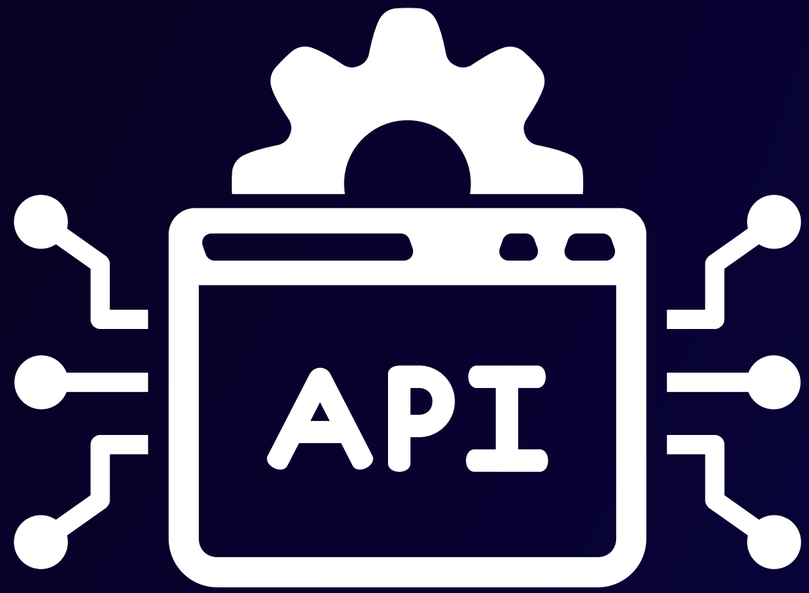


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OpenAPI

OpenAPI Specification (OAS)—formerly known as Swagger—is a universally accepted, vendor-neutral standard for describing RESTful HTTP APIs. Written in JSON or YAML, it creates a machine-readable contract of your API's endpoints, parameters, and authentication methods.

OPENAPI INITIATIVE



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OpenAPI – Struktura dokumentacji

paths	the URL patterns exposed by the API, e.g. /tasks/{id}
operation	a single combination of an HTTP method and a path, e.g. PUT /tasks/{id}
parameters	inputs passed via the URL or headers (path, query, header)
requestBody	the data sent in the body of the request
responses	the possible outcomes of an operation, grouped by status code





Termin	Znaczenie	Tłumaczenie
`path parameter`	a variable part of the URL path, e.g. {id} in /tasks/{id}	parametr ścieżki
`query parameter`	a value appended to a URL after "?" to filter or modify a request	parametr zapytania
`request body`	the data sent by the client in the body of the request	treść żądania
`schema`	a formal description of the structure and types of a piece of data	schemat danych
`required / optional`	whether a field must be present, or may be omitted	wymagane / opcjonalne
`enum`	a fixed set of allowed values for a field	zbiór dozwolonych wartości
`operation`	a single combination of an HTTP method and a path, e.g. POST /tasks	operacja (metoda + ścieżka)
`authentication token`	a piece of data proving the identity of the caller	token uwierzytelniający



Gdzie podróżują parametry?

path parameter

```
/tasks/{id}
```

part of the URL itself — identifies a specific resource

query parameter

```
?status=open
```

appended after "?" — filters or modifies the request

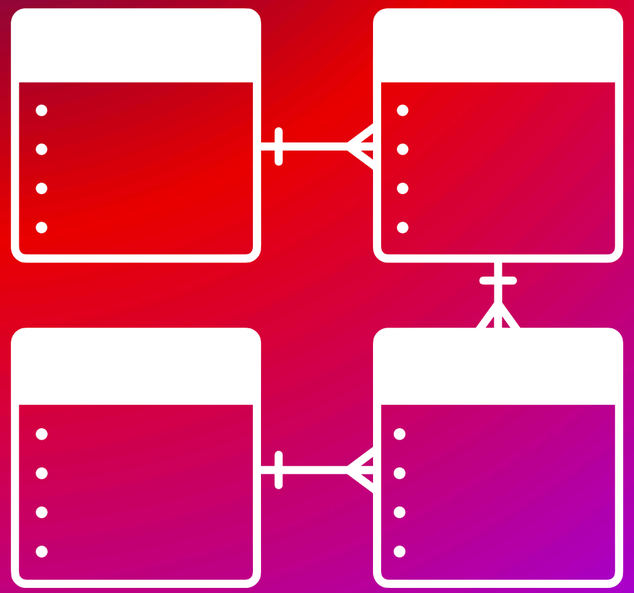
header

```
Authorization: Bearer ...
```

metadata sent alongside the request, not in the URL

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schema



a formal description of the structure and types of a piece of data

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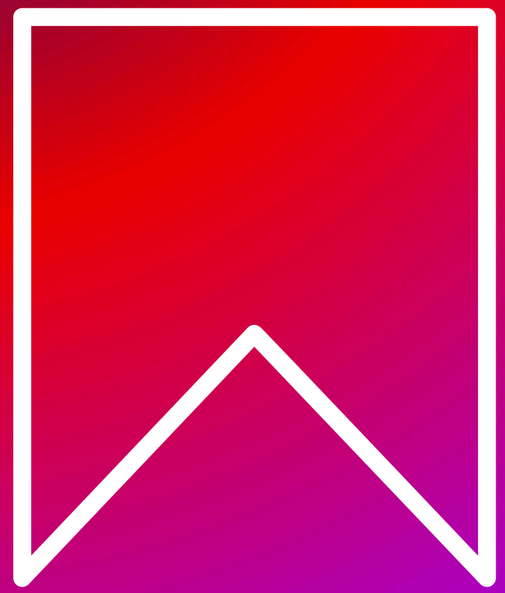
required / optional



whether a field must be present, or may be omitted

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enum



a fixed set of allowed values for a field,
e.g. "open" | "in_progress" | "done"

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Exercise – endpoint

```
POST /api/tasks/{id}/comments
Summary: Add a comment to an existing task.

Parameters:
  id            (path, string, required)   – the task's unique identifier
  Authorization (header, string, required) – Bearer token

Request body (application/json):
  text       (string, required)  – the comment content
  visibility (string, optional)  – one of: "team", "private"
                                   (default: "team")

Responses:
  201 – comment created; returns the comment object
  400 – the text field is missing or empty
  404 – no task exists with the given id
```

1. **Which HTTP method does this operation use?**
2. **Where is the id parameter located — path, query, or header?**
3. **Which THREE values is the status field allowed to have?**
4. **Which status code is returned if the Authorization header is missing?**
5. **Which status code is returned if the task does not exist?**

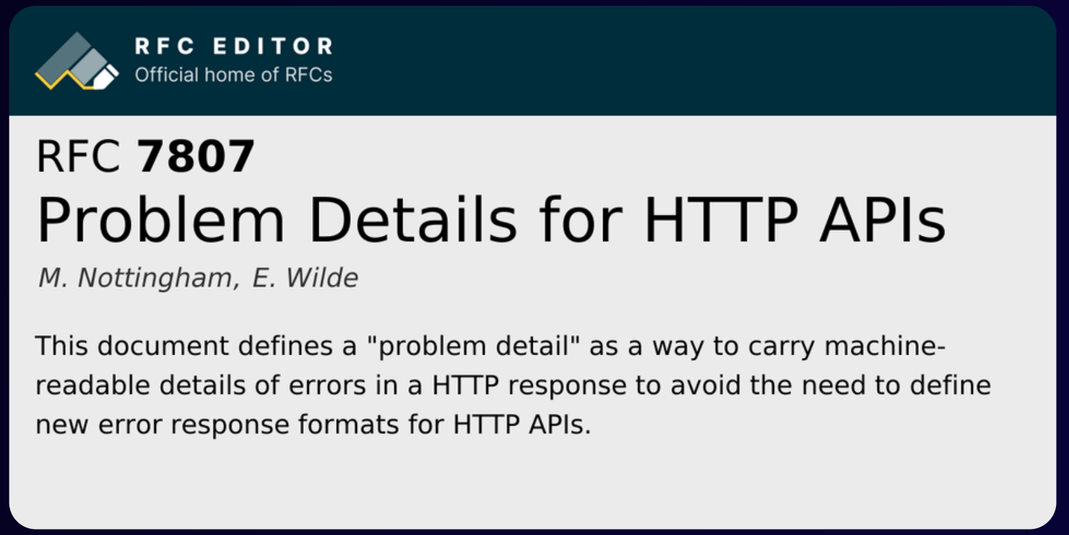
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RFC 7807

RFC 7807 is a technical standard created by the Internet Engineering Task Force (IETF) that defines a standardized, machine-readable format for returning error details from HTTP APIs.







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RFC 7807 – pola obiektu błędu

Pole	Znaczenie
`type`	a URI that identifies the problem type (ideally links to documentation about it)
`title`	a short, human-readable summary of the problem type — stays the same for the same type
`status`	the HTTP status code for this occurrence of the problem
`detail`	a human-readable explanation specific to this occurrence of the problem
`instance`	a URI identifying this specific occurrence of the problem





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RFC 7807 – pola obiektu błędu

```
HTTP/1.1 400 Bad Request
Content-Type: application/problem+json

{
  "type": "https://api.taskflow.io/errors/invalid-status",
  "title": "Invalid Status Value",
  "status": 400,
  "detail": "The status field must be one of: open, in_progress, done. Received: archived.",
  "instance": "/api/tasks/482/status"
}
```




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How to write good error messages?

Be specific	name the exact field or condition that caused the problem, not just "error" or "invalid".
Be constructive	tell the user what to do next, not only what went wrong.
Avoid blaming the user	describe the condition ("the token is missing") rather than accusing ("you forgot the token").
Avoid vague, generic words on their own	"error", "failed", "invalid", "unexpected error".

Use "please" only when asking the user to do something inconvenient – not for standard actions



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Exercise – error messages

Zły komunikat	Problem	Poprawiona wersja
"Error 500. Something went wrong."		
"Invalid input."		
"ERROR!!! Bad request."		
"You entered the wrong value."		

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APLIKACJE WEBOWE

:wq

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